Fixed Heavy-Tailed Probabilistic Forecasting under Missing Historical Observations

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Running title: Heavy-Tailed Forecasts with Missingness

Abstract

Forecasting systems often require predictive uncertainty as well as point forecasts, yet incomplete historical observations change the error distribution used to estimate that uncertainty. Comparisons are additionally weakened when competing methods see different missingness realizations. We study probabilistic time-series forecasting under fixed masks applied only to historical inputs, with train-only normalization and paired evaluation assets. A lightweight distribution head represents each horizon–channel marginal with a fixed Student-t distribution with degrees of freedom 8, using the same trainable location and scale projections as Gaussian and Laplace alternatives. Proper probabilistic scores are central: negative log-likelihood and a Monte Carlo estimate of the continuous ranked probability score are reported alongside point-prediction noninferiority. On fixed TimePerceiver checkpoints, Student-t improves negative log-likelihood in 12/12 dataset–seed pairs and is noninferior for both continuous ranked probability score and point prediction in 12/12. A distribution-head ablation supports fixed heavy tails over learned degrees of freedom in the evaluated protocol. On PatchTST, probabilistic-score gains appear in 3/3 datasets, while Electricity exceeds the prespecified point-error tolerance. The results support fixed heavy-tailed predictive distributions as a lightweight method for improving probabilistic forecast quality under missing historical observations, while the PatchTST Electricity result cautions against assuming metric-wise invariance across backbones.

Keywords: probabilistic forecasting; time-series forecasting; missing observations; Student-t distribution; proper scoring rules; heavy-tailed errors

1 Introduction

Forecasting decisions require more than a point estimate when the costs of under- and over-prediction differ. Predictive distributions express uncertainty across the forecast horizon and support decisions that depend on tail risk, quantiles, or intervals. Modern probabilistic forecasting therefore evaluates the entire distribution, not only its center (Gneiting and Raftery, 2007, Gneiting and Katzfuss, 2014, Salinas et al., 2020).

Historical observations are often incomplete. Missing values can remove recent evidence, entire channels, or contiguous blocks, changing both point errors and the residual shape available to a probability model. Recurrent imputation methods, deep generative imputers, and masked forecasting architectures address different parts of this problem (Cao et al., 2018, Che et al., 2018,

Fortuin et al., 2020, Tashiro et al., 2021, Du et al., 2023, Lee and Lee, 2025), but Gaussian residual assumptions can remain fragile when errors become heavy-tailed.

Probabilistic forecasts should be compared with proper scores that reward honest distributions. Negative log-likelihood (NLL) is distribution-sensitive, while the continuous ranked probability score (CRPS) evaluates calibration and sharpness together (Matheson and Winkler, 1976, Gneiting and Raftery, 2007, Jordan et al., 2019). Mean squared error (MSE) and mean absolute error (MAE) remain useful point criteria, but neither replaces distributional evaluation. Coverage and width are likewise diagnostics rather than complete quality measures in isolation.

Paired comparisons also require identical evaluation inputs. If Gaussian and heavy-tailed heads are evaluated under independently sampled masks, mask variation can be mistaken for a distribution-head effect. We therefore fix train, validation, and test splits; fit normalization on training data only; and reuse the same validation and test masks within every comparison.

The proposed final specification is deliberately small: a fixed Student-t predictive marginal with $\mathrm{df} = 8$. Gaussian, Laplace, and fixed Student-t variants have identical trainable location and scale projections; the fixed head adds no trainable degrees-of-freedom parameter. The value 8 is a reference setting that is empirically sufficient in the distribution-head ablation, not a theoretically optimal or dataset-specific choice.

The evidence combines missing-history degradation, four probability-head seeds on fixed TimePerceiver checkpoints, a distribution-family ablation, a PatchTST replication, efficiency measurements, and retained negative results. Our contributions are: (i) a fixed-mask protocol for paired probabilistic forecasting under missing historical observations; (ii) controlled evidence that a matched-capacity fixed heavy-tailed head improves proper scores on the evaluated TimePerceiver conditions; and (iii) a transparent second-backbone assessment showing that probabilistic-score gains transfer while a small Electricity point-error tradeoff prevents a complete multi-metric conclusion.

2 Related Work

2.1 Probabilistic forecasting and proper scores

DeepAR established autoregressive probabilistic forecasting at scale (Salinas et al., 2020); spline quantile functions and diffusion-based TimeGrad broadened the available forecast distributions (Gasthaus et al., 2019, Rasul et al., 2021). Proper-score methodology distinguishes distributional quality from point accuracy (Matheson and Winkler, 1976, Gneiting and Raftery, 2007, Gneiting et al., 2007, Gneiting and Katzfuss, 2014). We report NLL and a sampled CRPS implementation, with interval scores and coverage–width diagnostics as complementary summaries (Winkler, 1972, Jordan et al., 2019).

2.2 Heavy-tailed predictive distributions

The Student distribution originates from small-sample inference (Student, 1908) and has long been used for robust modeling because heavier tails reduce the penalty assigned to extreme residuals relative to a Gaussian model (Lange et al., 1989). Our use is narrower: we replace only the marginal forecast likelihood while keeping trainable location and scale capacity matched. The work neither introduces a new distribution nor learns a separate tail parameter for the final method.

2.3 Forecasting with incomplete histories

GRU-D, BRITS, GP-VAE, CSDI, and SAITS represent influential approaches to irregular or missing time-series data (Che et al., 2018, Cao et al., 2018, Fortuin et al., 2020, Tashiro et al., 2021, Du et al., 2023). TimePerceiver directly motivates the backbone and missing-history setting studied here (Lee and Lee, 2025). Our focus differs from imputation-only evaluation: future targets remain intact, and the forecast distribution is scored after fixed historical masking.

2.4 Forecasting backbones, calibration, and comparison

Informer, Autoformer, FEDformer, the Temporal Fusion Transformer, and PatchTST illustrate diverse sequence-modeling choices for long-horizon forecasting (Zhou et al., 2021, Wu et al., 2021, Zhou et al., 2022, Lim et al., 2021, Nie et al., 2023). Post-hoc probabilistic calibration and conformal intervals offer complementary tools (Kuleshov et al., 2018, Romano et al., 2019, Stankeviciute et al., 2021), although dependent time series require careful temporal splits. Finally, paired forecast comparisons should separate descriptive consistency from formal inferential testing (Diebold and Mariano, 1995, Giacomini and White, 2006); the present study reports no formal significance test.

3 Forecasting Problem

Let $x_{1:t} \in \mathbb{R}^{t \times C}$ denote a C-channel historical series observed up to forecast origin t, and let $y_{t+1:t+H}$ be the future target over horizon H. A binary historical observation mask $m_{1:t}$ marks available values. The model receives the masked historical input $\widetilde{x}_{1:t}$ together with the time covariates required by the backbone. Only historical inputs are masked; future targets are never masked or exposed to the model.

For every horizon h and channel c, the probabilistic head estimates a predictive marginal distribution

$$p(y_{t+h,c} \mid \widetilde{x}_{1:t}, m_{1:t}). \tag{1}$$

The point forecast is the predicted location. Central marginal prediction intervals are obtained from the corresponding distribution quantiles. These are horizon–channel marginals: the method does not estimate a joint covariance structure, a multivariate predictive density, or joint coverage.

The official chronological train, validation, and test splits are retained. Normalization statistics are fitted on training data only and reused for validation and test. Fixed mask assets define random, block, channel, and mixed missingness at rates 0.1, 0.3, 0.5, and 0.7 for the deterministic audit. Probability-head comparisons use paired fixed masks, preventing mask resampling from confounding distribution-family differences.

4 Probabilistic Forecasting Method

4.1 Matched-capacity marginal heads

Given a frozen backbone representation $z_{h,c}$, two linear projections produce location $\mu_{h,c}$ and a positive scale $\sigma_{h,c} = \mathrm{softplus}(s_{h,c}) + \epsilon$. Gaussian, Laplace, and fixed Student-t heads use exactly these trainable projections. The fixed head is

$$y_{t+h,c} \sim \mathrm{StudentT}(\mathrm{df} = 8, \mu_{h,c}, \sigma_{h,c}). \tag{2}$$

Thus it adds no trainable degrees-of-freedom parameters. A learned-df version is included only as an ablation and is not the proposed final method.

The fixed value df = 8 is a reference degrees-of-freedom setting. It is empirically sufficient under the evaluated distribution-head protocol, not theoretically optimal, and was not selected separately for each dataset. One goal of the ablation is to ask whether extra tail flexibility is supported once location and scale capacity are controlled.

4.2 Training objective and forecast summaries

Probability heads minimize validation-selected NLL while the official backbone checkpoint remains fixed. No test observation or target is used for head selection. NLL is a distribution-sensitive proper score. CRPS is another proper score that combines calibration and sharpness; here it is estimated with 64 Monte Carlo samples using the fixed evaluation seed 52025, rather than reported as an exact analytic value. MSE and MAE evaluate the location forecast.

For marginal intervals, prediction interval coverage probability (PICP) and mean prediction interval width (MPIW) describe coverage and sharpness separately. Neither PICP alone nor width alone is sufficient. The interval score combines coverage penalties with width for each marginal interval (Winkler, 1972). These diagnostics do not imply joint calibration.

5 Evaluation Design

5.1 Datasets and fixed missing-history protocol

Experiments use ETTh1, Weather, and Electricity with input length 384 and prediction length 96. The official splits and preprocessing rules are preserved, normalization is fitted on training data only, and fixed validation and test masks are reused within paired comparisons. Historical inputs alone are masked. The test set is never used for training, hyperparameter selection, early stopping, or conformal calibration.

The deterministic audit covers four mask types and four rates. The probabilistic TimePerceiver and PatchTST comparisons use the corrected fixed-mask pipeline; invalid historical runs in which masks were not applied are excluded from every result. Head selection uses validation NLL. Gaussian and Student-t heads within a pair use the same backbone checkpoint, mask, split, target, training budget, and evaluation code.

5.2 Seed scope and decision criteria

The TimePerceiver comparison uses four probability-head training seeds (2021, 2022, 2023, and 2025) on a fixed official backbone checkpoint per dataset. It is therefore a probability-head seed study, not an end-to-end backbone retraining study. The PatchTST comparison uses one backbone seed and one probability-head seed (2025) per dataset under random missingness at rate 0.3.

The paired criteria require lower NLL, CRPS noninferiority, and point-prediction noninferiority. Point noninferiority permits a relative MSE increase of at most 1%:

$$\frac{\mathrm{MSE}_t - \mathrm{MSE}_g}{\mathrm{MSE}_g} \leq 0.01, \tag{3}$$

where subscripts t and g denote fixed Student-t and Gaussian heads. This tolerance was fixed

Table 1: Deterministic forecast degradation under fixed historical masks. MSE is lower-is-better; relative degradation uses the corresponding clean TimePerceiver forecast as reference.

Dataset	Mask	Rate	MSE	Clean degradation
ETTh1	Block	0.30	0.423837	13.19%
ETTh1	Channel	0.30	0.583704	55.88%
ETTh1	Mixed	0.30	0.495055	32.20%
ETTh1	Random	0.10	0.378811	1.16%
ETTh1	Random	0.30	0.425394	13.60%
ETTh1	Random	0.50	0.525856	40.43%
Electricity	Block	0.30	0.184630	47.28%
Electricity	Channel	0.30	0.391038	211.92%
Electricity	Mixed	0.30	0.246785	96.86%
Electricity	Random	0.10	0.138429	10.42%
Electricity	Random	0.30	0.200572	59.99%
Electricity	Random	0.50	0.325732	159.83%
Weather	Block	0.30	0.180420	25.24%
Weather	Channel	0.30	0.284471	97.47%
Weather	Mixed	0.30	0.224939	56.14%
Weather	Random	0.10	0.169631	17.75%
Weather	Random	0.30	0.240810	67.16%
Weather	Random	0.50	0.331313	129.99%

before the second-backbone decision. Test results do not alter it. All reported comparisons are descriptive; no formal significance test is performed.

6 Empirical Results

6.1 Forecast degradation under missing history

Across 48 dataset–mask–rate cells, deterministic error rises as more historical values are removed. Table 1 gives the representative matrix and Figure 1 shows the complete rate trends; full tables appear in the supplement.

6.2 TimePerceiver paired proper-score results

Across 12 dataset–seed pairs, the Student-t variant improves NLL in 12/12 cases, is CRPS-noninferior in 12/12 cases, and satisfies point noninferiority in 12/12 cases. These are four

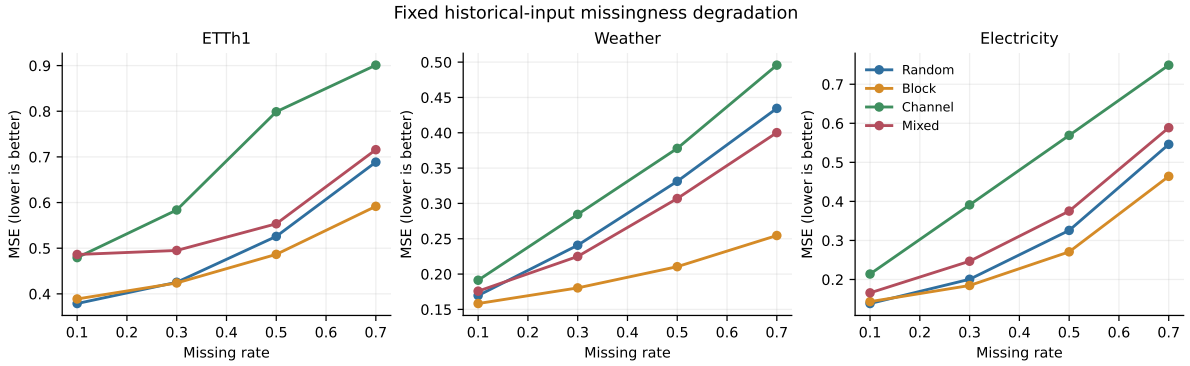


Figure 1: MSE under fixed historical-mask assets. Line styles and markers distinguish mask types in addition to color, and larger missing rates generally increase forecast error.

Table 2: TimePerceiver results across four probability-head seeds on fixed backbone checkpoints. Deltas are fixed Student-t minus Gaussian; negative NLL and CRPS deltas indicate improvement.

Dataset	G NLL	T NLL	ΔNLL	NLL wins	G CRPS	T CRPS	ΔCRPS	G MSE	T MSE	ΔMSE	Point NI
ETTh1	0.915014	0.871113	-0.043900	4	0.342011	0.337656	-0.004356	0.416862	0.415611	-0.001252	4
Weather	0.577226	0.103670	-0.473557	4	0.244788	0.207544	-0.037245	0.210306	0.209818	-0.000488	4
Electricity	0.439644	0.365348	-0.074296	4	0.209365	0.207395	-0.001970	0.167419	0.167703	0.000284	4

probability-head seeds on a fixed backbone checkpoint for each dataset. Table 2 reports dataset means and variability; Figure 2 retains every paired NLL delta.

6.3 Distribution-family ablation

The matched-head ablation compares Gaussian, Laplace, fixed Student-t, and learned-df Student-t forecasts. Table 3 and Figure 3 support fixed heavy-tailed forecasting rather than learned degrees of freedom. Fixed df $= 8$ is sufficient under the evaluated protocol, whereas learning per-channel df provides no consistent additional benefit. The earlier learned-df results remain visible in the ablation.

6.4 PatchTST replication and point tradeoff

PatchTST records NLL wins in 3/3 datasets, CRPS noninferiority in 3/3, point noninferiority in 2/3, and a full paired pass in 2/3. ETTh1 and Weather pass all paired criteria. For Electricity, Gaussian MSE is 0.165829 and fixed Student-t MSE is 0.167809, a change of +0.001980 or +1.194%. This exceeds the prespecified 1% point-noninferiority threshold. Thus probabilistic-score gains transfer to the second evaluated backbone, but complete multi-metric consistency does not.

6.5 Efficiency

Fixed Student-t has the same number of trainable head parameters as Gaussian and Laplace because df is not learned. Table 5 reports controlled timing and memory measurements. Comparisons are interpreted within each benchmark environment, not as hardware-independent throughput claims.

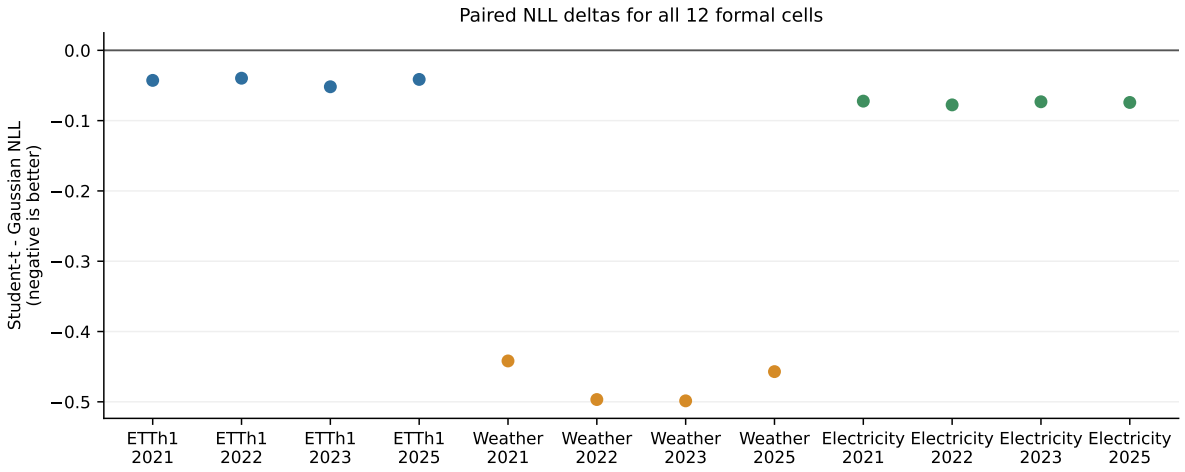


Figure 2: Paired TimePerceiver NLL deltas for all dataset–head-seed pairs. Negative values favor the fixed Student-t forecast distribution.

Table 3: Distribution-head ablation on frozen TimePerceiver checkpoints. Gaussian, Laplace, and fixed Student-t share identical trainable location and scale capacity; lower NLL, CRPS, MSE, and MAE are better.

Dataset	Head	NLL	CRPS	MSE	Coverage error	Trainable
etth1	Gaussian	0.918535	0.344127	0.420332	0.038132	6,168
etth1	Laplace	0.879116	0.341785	0.424149	0.011910	6,168
etth1	Fixed Student-t (df=8)	0.864949	0.336548	0.413927	0.022006	6,168
etth1	Learned Student-t	0.877154	0.340263	0.419177	0.020301	6,175
weather	Gaussian	0.586325	0.245442	0.212237	0.082667	24,624
weather	Laplace	0.204399	0.209589	0.210051	0.070806	24,624
weather	Fixed Student-t (df=8)	0.136165	0.208461	0.210994	0.051632	24,624
weather	Learned Student-t	0.129293	0.208276	0.210836	0.052156	24,645
electricity	Gaussian	0.440598	0.209453	0.167881	0.038304	24,624
electricity	Laplace	0.370388	0.208768	0.170436	0.018222	24,624
electricity	Fixed Student-t (df=8)	0.367009	0.207797	0.168473	0.018744	24,624
electricity	Learned Student-t	0.366449	0.207786	0.168486	0.018502	24,945

6.6 Tail-concentrated gains

Residual-percentile decompositions place much of the NLL improvement in the most extreme Gaussian-residual bins. Figure 5 presents both evaluated backbones. These diagnostics are descriptive and do not establish a causal mechanism or treat forecast elements as independent inferential samples.

7 Diagnostics and Negative Results

The retained negative results constrain the interpretation. On ETTh1 with random missingness at rate 0.3, a matched-capacity static adapter improves the official forecast, but the tested missingness-aware FiLM adapter does not improve over that static adapter. Mechanism checks show nonconstant conditioning, nonzero FiLM modulation, finite gradients, and parameter updates. The result

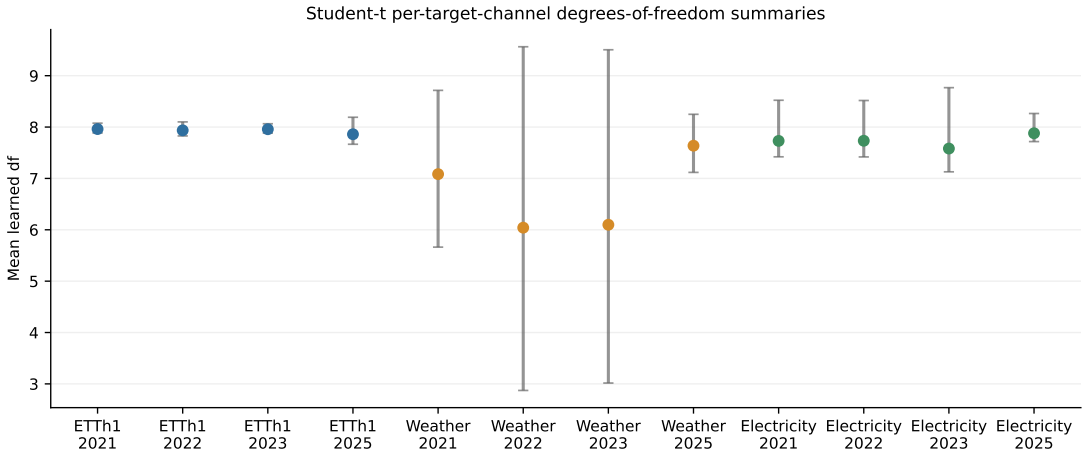


Figure 3: Distribution-head comparison under random historical missingness at rate 0.3. The fixed df = 8 setting is a common reference value, not a dataset-specific selection.

Table 4: PatchTST fixed Student-t minus Gaussian deltas. Negative NLL and CRPS values indicate improvement. Electricity's +1.194% relative MSE change exceeds the 1% point-noninferiority threshold.

Dataset	Backbone	ΔNLL	ΔCRPS	ΔMSE	Rel. MSE	NLL win	CRPS NI	Point NI	Full pair
ETTh1	TimePerceiver	-0.053587	-0.007578	-0.006405	-1.524%	Yes	Yes	Yes	Yes
ETTh1	PatchTST	-0.035317	-0.002538	-0.005439	-1.274%	Yes	Yes	Yes	Yes
Weather	TimePerceiver	-0.450160	-0.036981	-0.001243	-0.586%	Yes	Yes	Yes	Yes
Weather	PatchTST	-1.818698	-0.076205	-0.050179	-21.595%	Yes	Yes	Yes	Yes
Electricity	TimePerceiver	-0.073589	-0.001656	0.000592	+0.353%	Yes	Yes	Yes	Yes
Electricity	PatchTST	-0.083433	-0.001688	0.001980	+1.194%	Yes	Yes	No	No

therefore concerns this global-statistics conditioning design and fixed condition; it does not establish that missingness conditioning is generally unhelpful.

Under a strict raw-time selection and calibration split, the evaluated additive split-conformal correction widens intervals without improving absolute coverage error. Native intervals are already near the nominal targets, while the calibration segment is descriptively harder. This is a result for one additive score, split, dataset, and missingness condition; it does not generalize to other conformal designs. Detailed tables and figures for both negative findings are in the supplement.

8 Discussion

8.1 Why heavy tails help

The residual-bin analyses show that NLL gains are concentrated in extreme Gaussian-residual regions on both evaluated backbones. This pattern is consistent with a heavier-tailed forecast assigning less implausibly small density to large errors. It is descriptive, not causal proof: the experiments do not manipulate tail behavior independently of the likelihood family.

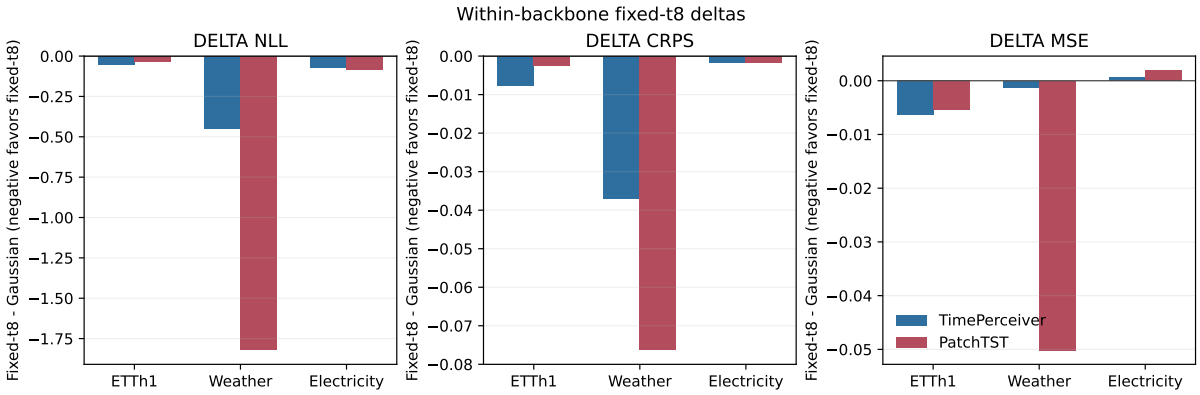


Figure 4: Within-backbone metric deltas for TimePerceiver and PatchTST. Negative values favor fixed Student-t; the positive PatchTST Electricity MSE delta is retained.

Table 5: Controlled probability-head parameter, runtime, and memory measurements. Values are compared within dataset and benchmark environment.

Dataset	Head	Params	Forward ms	NLL ms	CRPS ms	Peak MiB
etth1	Gaussian	6,168	29.997997	51.514500	86.934799	215.033691
etth1	Laplace	6,168	29.996887	30.016349	63.189127	215.033691
etth1	Fixed Student-t (df=8)	6,168	29.902914	65.900001	90.256716	215.034180
etth1	Learned Student-t	6,175	29.982518	69.371223	103.335362	215.034180
weather	Gaussian	24,624	59.357760	79.919755	188.685383	630.664551
weather	Laplace	24,624	58.485584	59.293318	133.341993	630.664551
weather	Fixed Student-t (df=8)	24,624	58.181018	98.037041	211.123562	630.665039
weather	Learned Student-t	24,645	59.026097	97.896011	212.516092	630.665039
electricity	Gaussian	24,624	356.894562	359.007118	1248.479182	4626.902832
electricity	Laplace	24,624	345.290164	343.346301	895.849843	4626.902832
electricity	Fixed Student-t (df=8)	24,624	343.877731	374.742924	1477.667372	4626.904297
electricity	Learned Student-t	24,945	420.384170	473.496327	1890.070674	4626.904297

8.2 Why a fixed degree of freedom can suffice

A fixed df provides robust tails without adding learned shape parameters. One possible explanation for the learned-df result is that per-channel flexibility may add estimation variance or be weakly identified under the available head-training data. This remains a hypothesis. The evidence only shows that learned df did not yield a consistent additional gain in this ablation.

8.3 Proper-score versus point tradeoff

The PatchTST Electricity case illustrates why distributional and point objectives must be reported together. Better NLL and CRPS can coexist with a small MSE deterioration because the proper scores evaluate the predictive distribution while MSE evaluates its center. The +1.194% deterioration exceeds the fixed 1% tolerance, so it is a genuine protocol-level tradeoff rather than a rounding artifact.

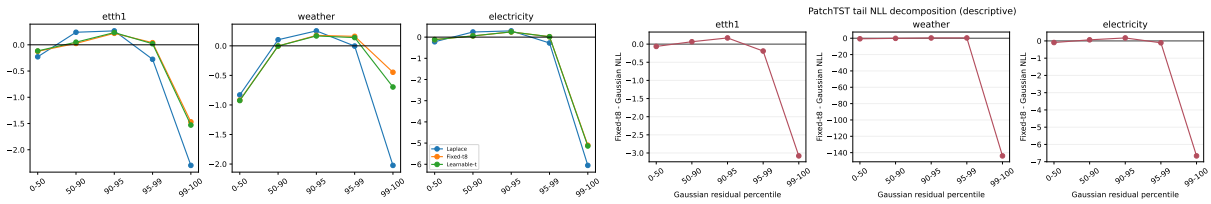


Figure 5: Descriptive NLL differences across Gaussian-residual percentile bins for TimePerceiver (left) and PatchTST (right). Negative values favor fixed Student-t.

8.4 Practical interpretation

The supported use case is a forecasting system that requires marginal predictive uncertainty under incomplete historical inputs and can retain paired fixed-mask evaluation. Fixed Student-t is attractive here because it changes the likelihood without increasing trainable head capacity. This interpretation does not imply a particular business deployment, joint uncertainty, or invariant point performance across architectures.

9 Limitations

The evidence covers three datasets and two backbones with fixed historical masks. Some conditions are single-seed. TimePerceiver varies four probability-head seeds while holding the backbone checkpoint fixed, and PatchTST uses one backbone seed and one head seed under random missingness at rate 0.3 only. PatchTST Electricity has a point-prediction tradeoff. No formal significance test is reported.

The predictive distributions are marginal by horizon and channel. We do not model joint covariance, assess joint coverage, or report a multivariate energy score. The fixed df $= 8$ value is not theoretically optimal, and the study does not perform dataset-specific tail selection. The conformal study covers a limited additive split procedure, and the FiLM study covers a limited global conditioning design.

The experiments do not evaluate decision-specific economic loss, domain-specific forecast value, density forecast combination, or operational consequences. These are untested extensions rather than implied benefits. Broader missingness shifts, more independently trained backbones, and alternative tail models are needed before stronger generalization claims are warranted.

10 Conclusion

This study evaluates a matched-capacity fixed Student-t forecast head under fixed masks applied only to historical observations. Proper probabilistic scores improve consistently across the evaluated TimePerceiver probability-head seeds, and a distribution ablation supports fixed heavy tails over learned degrees of freedom in this setting. PatchTST reproduces the NLL and CRPS direction on all three datasets but retains a small Electricity MSE tradeoff that exceeds the prespecified tolerance. The resulting conclusion is deliberately bounded: fixed heavy-tailed predictive marginals are a lightweight way to improve probabilistic forecast quality under the evaluated missing-history protocol, but point behavior and broader generalization must be checked for each backbone and condition.

Statements and Declarations

Data availability. The datasets are publicly available from their original repositories; sources and preprocessing details are documented in the supplementary material.
Code availability. [AUTHOR ACTION REQUIRED: provide the verified repository URL and release timing.]
Funding, conflicts, and contributions. [AUTHOR INPUT REQUIRED.]
Generative AI. [AUTHOR CONFIRMATION REQUIRED: see the separate disclosure draft.]

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